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RESEARCH INTEREST

Soil physics; Poromechanics; Phase transitions in porous media.

EDUCATION

Ph.D., Geotechnical Engineering Hunan University, Changsha, China Supervisor: Prof. Chao Zhang	Sep. 2020 – Present
B.S., Civil Engineering Hunan University, Changsha, China	Sep. 2016 – Jun. 2020

PUBLICATIONS

Journal Articles (* Corresponding author)

- [1] **Hu, S.**, Zhang, C.*, Li, L., & Lu, N. (2026). Determining soil water structure, density, and viscosity via infrared spectroscopy. *Journal of Engineering Mechanics*, 152(4), 04026002. DOI: 10.1061/JENMDT.EMENG-8618
- [2] **Hu, S.**, Zhao, N., Zhang, C.*, Li, F., Chen, R., & Or, D.* (2025). Water nanofilms facilitate ice crystal growth across droplets. *Physical Review Letters*, 134(6), 064001. DOI: 10.1103/PhysRevLett.134.064001
- [3] **Hu, S.**, Zhang, C.*, Dong, Y., Gou, L., & Chen, R. (2025). Water vapor sorption isotherms of salt-affected soils. *Canadian Geotechnical Journal*, cgj-2024-0426. DOI: 10.1139/cgj-2024-0426
- [4] **Hu, S.**, Zhang, C.*, & Lu, N. (2023). Quantifying coupling effects between soil matric potential and osmotic potential. *Water Resources Research*, 59(2), e2022WR033779. DOI: 10.1029/2022WR033779
- [5] **Hu, S.**, & Zhang, C.* (2023). A Sorption Isotherm Model for Soil Incorporating External and Internal Surface Adsorption, and Capillarity. *Canadian Geotechnical Journal*, cgj-2022-0386. DOI: 10.1139/cgj-2022-0386
- [6] Zhang, C., Lin, X., **Hu, S.***, & Lu, N. (2025). Predicting the Soil Freezing Curve via a Modified Clapeyron Equation. *Journal of Geotechnical and Geoenvironmental Engineering*, 152(2), 04025195. DOI: 10.1061/JGGEFK.GTENG-13973
- [7] Zhang, C., Zhao, N., **Hu, S.***, & Lin, X. (2025). Salts retard ice crystal growth in supercooled droplets during recalescence. *Physical Review Fluids*, 10(6), 063605. DOI: 10.1103/mt89-z2jw
- [8] Gou, L., Lu, N., **Hu, S.**, Calderon, A. R. A., & Zhang, C.* (2025). Suction Stress of Soil Slurry. *Journal of Geotechnical and Geoenvironmental Engineering*, 151(2), 04024164. DOI: 10.1061/JGGEFK.GTENG-12758
- [9] Lin, X., Zhang, C.*, **Hu, S.**, & Chen, R. (2024). Heterogeneous ice nucleation of salt solution in porous media. *The Journal of Chemical Physics*, 160(9), 094501. DOI: 10.1063/5.0190862

- [10] Zhang, C., Li, L., **Hu, S.***, Gou, L., & Chen, R. (2024). Physical origin of adsorption heat and its significance in the isotherm equation. *International Journal of Heat and Mass Transfer*, 220, 124914. DOI: 10.1016/j.ijheatmasstransfer.2023.124914
- [11] Zhao, N., **Hu, S.**, Zhang, C.*, Li, F., & Chen, R. (2023). Physical Origins of Freezing and Melting Temperature Depressions of Water in Millimeter-sized Pores. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, 674, 131851. DOI: 10.1016/j.colsurfa.2023.131851
- [12] Gou, L., Zhang, C.*, Lu, N., & **Hu, S.** (2023). A Soil Hydraulic Conductivity Equation Incorporating Adsorption and Capillarity. *Journal of Geotechnical and Geoenvironmental Engineering*, 149(8), 04023056. DOI: 10.1061/JGGEFK.GTENG-11388
- [13] Gou, L., Zhang, C.*, **Hu, S.**, Chen, R., & Dong, Y. (2023). Semi-analytical Solutions for Soil Consolidation Induced by Drying. *Acta Geotechnica*, 18(2), 739–755. DOI: 10.1007/s11440-022-01623-4
- [14] Zhang, C., **Hu, S.**, Qiu, Z., & Lu, N.* (2022). A Poroelasticity Theory for Soil Incorporating Adsorption and Capillarity. *Géotechnique*, 1–18. DOI: 10.1680/jgeot.22.00097
- [15] Zhang, C., Gou, L., **Hu, S.***, & Lu, N. (2022). A Thermodynamic Formulation of Water Potential in Soil. *Water Resources Research*, 58(9). DOI: 10.1029/2022WR032369
- [16] Zhang, C.*, **Hu, S.**, & Lu, N. (2022). Unified Elastic Modulus Characteristic Curve Equation for Variably Saturated Soils. *Journal of Geotechnical and Geoenvironmental Engineering*, 148(1), 04021171. DOI: 10.1061/(ASCE)GT.1943-5606.0002718

PRESENTATIONS

2025 Annual Academic Conference of Chinese Society of Soil Science , Yantai, China	August 2025
Poster: Phase equilibrium and transitions in salt-affected soils	
Ice at the Micro-scale 2025 , Ascona, Switzerland	July 2025
Poster: Salts retard ice crystal growth in supercooled droplets during recalescence	
AGU24 , Washington, D.C., United States	December 2024
Poster: Phase equilibrium and transitions in salt-affected soils	
InterPore2024 , Qingdao, China	May 2024
Oral: Adsorption-induced Effective Stress in Porous Media (On behalf of Prof. Zhang)	
The 4th International Soil Modeling Consortium (ISMC) Conference , Tianjin, China	May 2024
Poster: Coupling Between Soil Matric Potential and Osmotic Potential	
InterPore2023 , Edinburgh, Scotland	May 2023
Poster: A Poroelasticity Theory for Soil Incorporating Adsorption and Capillarity	
The 9th Young Experts Forum on Geotechnical Engineering , Changsha, China	June 2021
Oral: Unified Elastic Modulus Function for Variably Saturated Soils	

ACADEMIC SERVICE

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Vadose Zone Journal, Associate Editor, 2026 - present
Vadose Zone Journal, Social Media Editor, 2025 - 2026

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